

AI ASSISTED CODING ASSIGNMENT 9

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TASK 1: FUNCTION

SUMMARY GENERATION

```
DEF CALCULATE_AREA(RADIUS):
```

```
    RETURN 3.14 *
```

```
RADIUS * RADIUS DEF
```

```
IS_EVEN(NUM):
```

```
    RETURN NUM
```

```
% 2 == 0 DEF
```

```
FACTORIAL(N):
```

```
IF N == 0:
```

```
    RETURN 1
```

```
    RETURN N * FACTORIAL(N-1)
```

AI RESPONSE (USED AS DOCSTRINGS):

```
DEF CALCULATE_AREA(RADIUS):
```

```
    """CALCULATES THE AREA OF A CIRCLE GIVEN
```

```

ITS RADIUS."""    RETURN 3.14 * RADIUS *
RADIUS DEF IS_EVEN(NUM):    """DETERMINES
IF A GIVEN NUMBER IS EVEN."""    RETURN NUM
% 2 == 0 DEF FACTORIAL(N):    """COMPUTES
THE FACTORIAL OF A NON-NEGATIVE
INTEGER."""    IF N == 0:    RETURN 1
RETURN N * FACTORIAL(N-1)

```

TASK 2: LOGICAL EXPLANATION FOR
CONDITIONS AND LOOPS ORIGINAL CODE
WITHOUT COMMENTS:

```

NUMBERS =
[1,2,3,4,5,6,7,8,9,10
] EVENS = [] FOR
NUM IN NUMBERS:
IF NUM % 2 == 0:

```

```
EVENS.APPEND(NUM  
)  
    ELSE:  
PRINT("ODD")  
PRINT(EVENS)
```

PROMPT:

EXPLAIN LOGIC BEHIND CONDITIONS AND LOOPS IN THIS CODE. ADD INLINE COMMENTS ONLY FOR DECISION LOGIC AND LOOP BEHAVIOR.

OUTPUT WITH EXPLANATIONS:

```
NUMBERS =
```

```
[1,2,3,4,5,6,7,8,9,10]
```

```
EVENS = []
```

```
FOR NUM IN NUMBERS: # LOOP ITERATES THROUGH  
    EACH NUMBER IN THE LIST TO CHECK PARITY    IF NUM  
    % 2 == 0: # CONDITION CHECKS IF NUMBER IS DIVISIBLE
```

```
BY 2 (EVEN)      EVENS.APPEND(NUM) # COLLECT
EVEN NUMBERS IN LIST  ELSE: # HANDLES ODD
NUMBERS

    PRINT("ODD") # OUTPUT FOR
ODD CASES PRINT(EVENS) #
DISPLAY FINAL LIST OF EVENS
```

TASK 3: FILE-LEVEL OVERVIEW

SAMPLE FILE CONTENT (MATH_UTILS.PY ORIGINAL):

```
DEF ADD(A,B):    RETURN A+B

DEF SUBTRACT(A,B):    RETURN
A-B DEF MULTIPLY(A,B):
RETURN A*B
```

PROMPT:

GENERATE HIGH-LEVEL OVERVIEW FOR THIS ENTIRE PYTHON FILE. SUMMARIZE PURPOSE AND FUNCTIONALITY. PLACE AT TOP.

AI GENERATED OVERVIEW ADDED:

```
"""
```

MATH UTILITIES MODULE THIS FILE PROVIDES BASIC ARITHMETIC OPERATIONS INCLUDING ADDITION, SUBTRACTION, AND MULTIPLICATION FOR NUMERICAL VALUES. USEFUL FOR SIMPLE CALCULATIONS IN MATH-HEAVY SCRIPTS.

```
""" DEF
```

```
ADD(A,B):
```

```
RETURN A+B
```

```
DEF
```

```
SUBTRACT(A,B):
```

```
    RETURN A-B
```

```
DEF
```

```
MULTIPLY(A,B):
```

```
    RETURN A*B
```

**TASK 4: REFINE EXISTING
DOCUMENTATION
ORIGINAL CODE WITH
POOR COMMENTS:**

```
# THIS DOES
```

```
SOMETHING DEF
```

```
DIVIDE(A, B):
```

```
    IF B != 0: #
```

```
CHECK
```

```
    RETURN A / B
```

```
ELSE:
```

```
PRINT("ERROR") #
```

```
BAD X = 10
```

```
FOR I IN
```

```
RANGE(5): #
```

```
LOOP    X += I #
```

```
END
```

PROMPT:

REFINE THESE COMMENTS FOR CLARITY AND
CONSISTENCY. KEEP MEANING SAME, IMPROVE
WORDING.

REFINED OUTPUT:

```
# DIVISION FUNCTION
```

```
WITH ZERO-CHECK DEF
```

```
DIVIDE(A, B):
```

```

"""DIVIDES A BY B, HANDLES DIVISION BY ZERO."""
IF B != 0: # ENSURES DENOMINATOR IS
NOT ZERO TO AVOID ERRORS      RETURN
A / B  ELSE:

    PRINT("CANNOT DIVIDE BY ZERO") #
ERROR MESSAGE FOR INVALID INPUT X = 10
FOR I IN RANGE(5): # ACCUMULATES SUM
OF FIRST 5 NUMBERS INTO X    X += I
PRINT(X) # OUTPUTS THE COMPUTED SUM

```

TASK 5: PROMPT DETAIL

IMPACT STUDY SAME

FUNCTION:

```

DEF MAX_IN_LIST(LST):  MAX_VAL
= LST[0]  FOR ITEM IN LST:
IF ITEM > MAX_VAL:

```



```
MAX_VAL = ITEM    RETURN  
MAX_VAL
```

BRIEF PROMPT:

ADD DOCSTRING TO THIS FUNCTION.

BRIEF OUTPUT:

```
DEF MAX_IN_LIST(LST):    """FINDS  
MAX IN LIST."""    MAX_VAL =  
LST[0]    FOR ITEM IN LST:    IF  
ITEM > MAX_VAL:        MAX_VAL =  
ITEM    RETURN MAX_VAL
```

DETAILED PROMPT:

WRITE A COMPREHENSIVE DOCSTRING FOR THIS
FUNCTION. INCLUDE: PURPOSE, PARAMETERS (TYPE AND
DESC), RETURNS (TYPE AND DESC), LOGIC EXPLANATION

WITHOUT CODE. USE GOOGLE STYLE IF POSSIBLE.
FUNCTION FINDS MAXIMUM VALUE IN A LIST USING
ITERATION. DEF MAX_IN_LIST(LST): [CODE]

DETAILED OUTPUT:

```
DEF MAX_IN_LIST(LST): """ FINDS  
THE MAXIMUM VALUE IN A NON-  
EMPTY LIST OF NUMBERS.
```

ARGS:

LST (LIST): A NON-EMPTY LIST OF
COMPARABLE NUMBERS.

RETURNS:

INT OR FLOAT: THE MAXIMUM
VALUE IN THE LIST. RAISES:

INDEXERROR: IF THE LIST IS EMPTY.

LOGIC: INITIALIZES MAX WITH FIRST ELEMENT,
ITERATES THROUGH LIST, UPDATES MAX IF LARGER
VALUE FOUND. """

```
MAX_VAL =  
LST[0]    FOR  
ITEM IN LST:  
    IF ITEM >  
MAX_VAL:  
    MAX_VAL = ITEM  
RETURN MAX_VAL
```

COMPARISON TABLE

ASPECT	BRIEF PROMPT OUTPUT	DETAILED PROMPT OUTPUT
COMPLETENES S	BASIC PURPOSE ONLY	FULL PARAMS, RETURNS, RAISES, LOGIC

CLARITY	VAGUE "FINDS MAX"	PRECISE DESCRIPTIONS, TYPES
ACCURACY	CORRECT BUT MINIMAL	ACCURATE, ADDS ERROR HANDLING NOTE
LENGTH	1 LINE	MULTI-LINE COMPREHENSIVE

